Fault Analysis of Microscaling Formats on a RISC-V SoC

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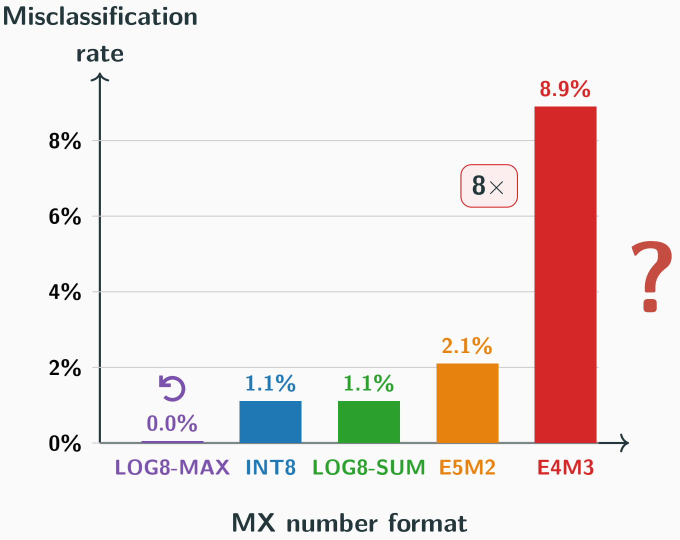
Same weights, same answer: why is one format 8× more vulnerable?



Four steps to the answer

1 — Why it matters

reliability & security at stake

2 — Background

5 formats · 2 fault levels

3 — Three mechanisms

amplify · invert · self-heal

4 — Results & takeaway

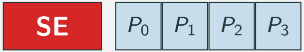
sim + silicon · 8× gap

Reliability of edge-AI inference on MX formats

- Edge-AI inference runs everywhere, on a tight memory / energy budget.
- **MX** (OCP 2023): one *shared* exponent per k=4 block; shipping in **NVIDIA, AMD, Intel**, RISC-V.
- In the field: **fault injection** (EMFI, glitch); reliability at stake.
- The **shared exponent** is a new fault surface prior work never examined.

First comparative MX fault study on a fabricated SoC.

MX block, k=4



ONE exponent shared by ALL four

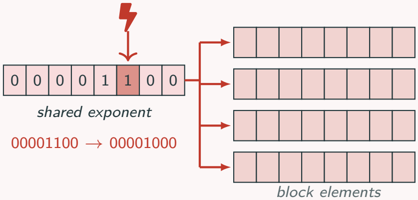
Five 8-bit MX formats in three families: INT, FP, LOG

	INT : linear	**FP : exponential**	**LOG : log-domain**
	S	S E E E E M M M	S FI FI FI FI FE FE FE
Range	$\pm$127, uniform	E4M3 $\pm$448 E5M2 $\pm$57344	up to $\sim 2^{16}$, log-spaced
Advantage	full accuracy	widest dynamic range	cheap multiplication
Variants	**MXINT8**	**MXFP8-E4M3**, **MXFP8-E5M2**	**LOG8-SUM**, **LOG8-MAX**
Sensitivity	**linear**	$2\times$ **to** $256\times$	$2\times$ **to** $256\times$

The key idea: faults act at two levels

LEVEL 1: shared-exponent fault

one flip rescales the whole block

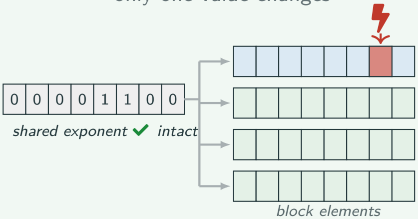


1 fault → all 4 values rescaled

$k\times$ fan-out, unique to MX

LEVEL 2: element fault

only one value changes



1 fault → only 1 value

When does a fault actually flip the answer?

A fault flips the answer only if it opens **both** locks.

fault → **Lock 1** — **Survive the reduction**: SUM keeps all terms, MAX keeps the winner → **Lock 2** — **Beat the margin**: shift beats the top-two gap → **silent misclassification**

golden scores

cat **9.0** winner

dog 8.4

bird 2.1

fish 1.0

margin 0.6

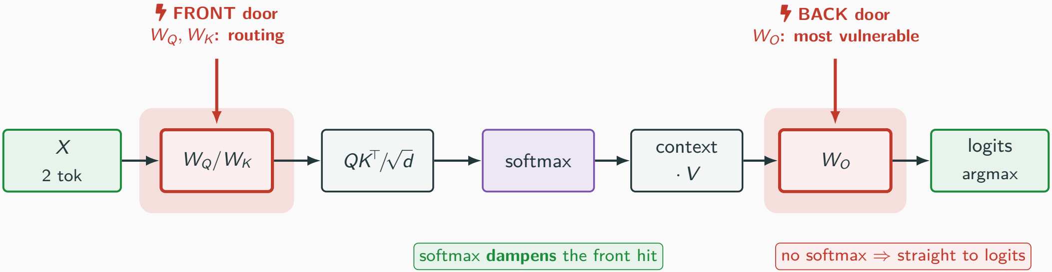
fault lifts a loser by 2^{Δ}

SUM ✖
passes Lock 1, then **flips**

MAX ✔
dropped at Lock 1, **no flip**

Inside attention: two doors, three mechanisms

Weights (W_Q, W_K, W_V, W_O) in **SRAM**; a fault may strike any, yet **two doors** flip the answer.



❶ SE amplification

BACK: $W_O \to$ logits, nothing absorbs it

❷ routing inversion

FRONT: W_Q/W_K reorder the softmax

❸ log self-healing

reduction: MAX drops the faulted loser

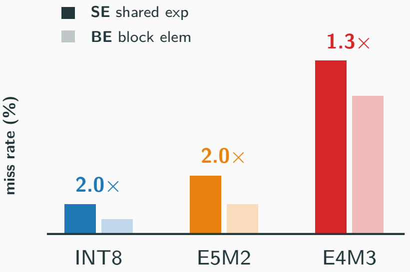
Mechanism 1: shared-exponent amplification

The shared exponent is one volume knob: flip one bit and all $k=4$ weights rescale at once.

One bit flips the shared exponent

	shared exp	*block elements ($k=4$)*				
golden	8	14	-8	4	10	
	↓ ⚡1 bit					
shared	$8 \to 16$	28	-16	8	20	**all 4** $\times 2^{\Delta}$ (fault factor)

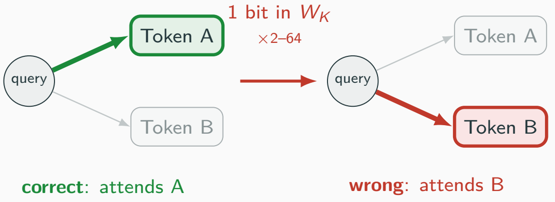
Amplification = SE-rate / BE-rate (per byte)



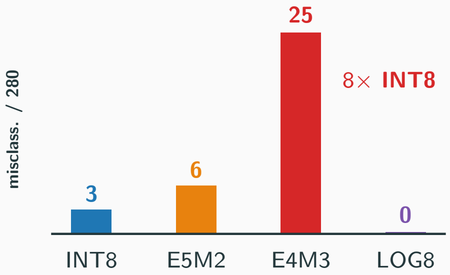
Mechanism 2: attention routing inversion

Routing inversion: a fault lifts a *loser's* score above the true winner, so softmax sends attention to the wrong token.

Fault swings attention to the wrong token



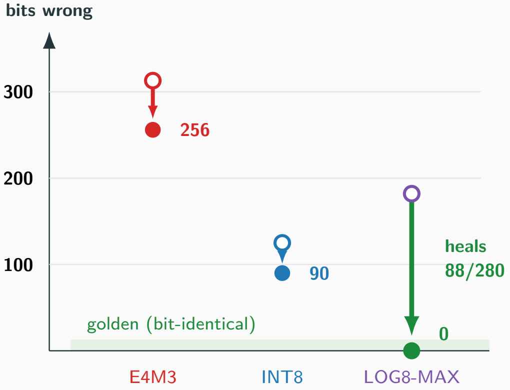
Wrong answers per 280 faults



Mechanism 3: log-domain self-healing

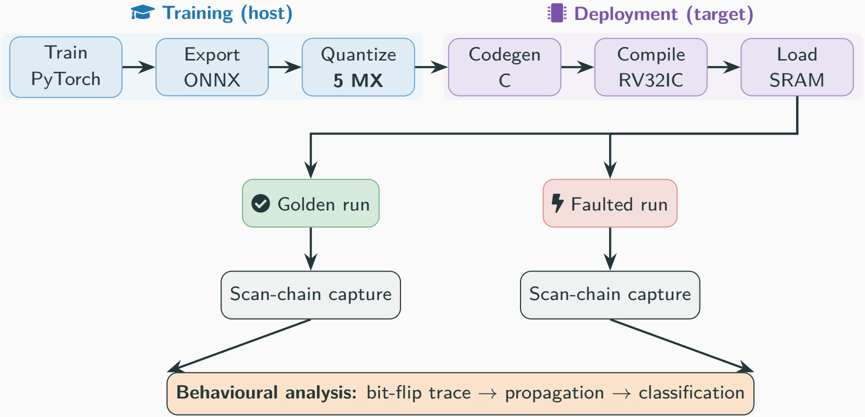
Self-healing: the faulted state converges back on its own to the **bit-exact** golden run, leaving **zero** residual error.

Peak (○), then where it settles (●)



- E4M3 and INT8 stay broken.
- LOG8-MAX heals back to **0**.
- Heals: **DROP** · **BOUND** · **WASH**.
- Fastest too: ∼2 vs ∼21 cyc/MAC.

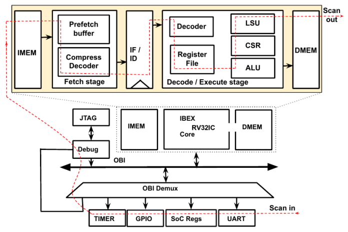
Method: one exhaustive fault-injection pipeline



Four architectures (self-attention, MoE attention, MLP, keyword spotting) are trained in PyTorch, quantized to all five MX formats, and run as on-chip **inference**.

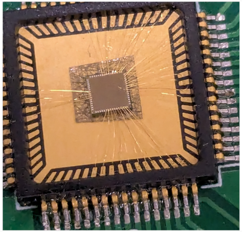
Platform: CAPRI1 in simulation and on silicon

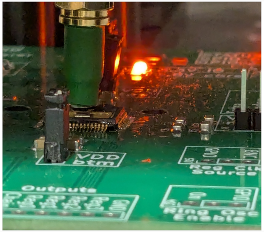
Simulation (gate-level)



- Netlist of the real SoC.
- **12,756-FF scan chain** captures full processor state every 50 cycles.

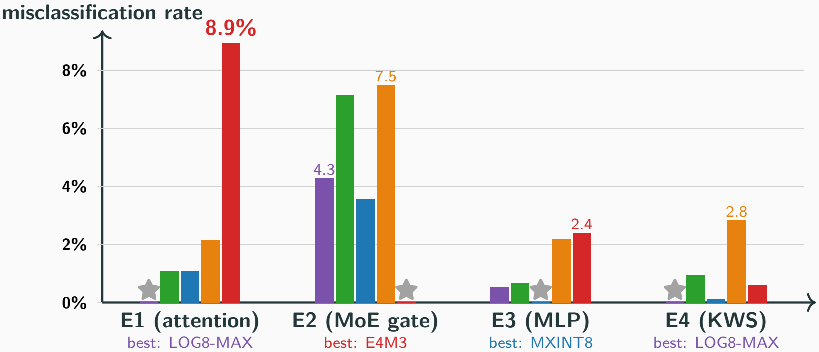
Measurement (silicon EMFI)





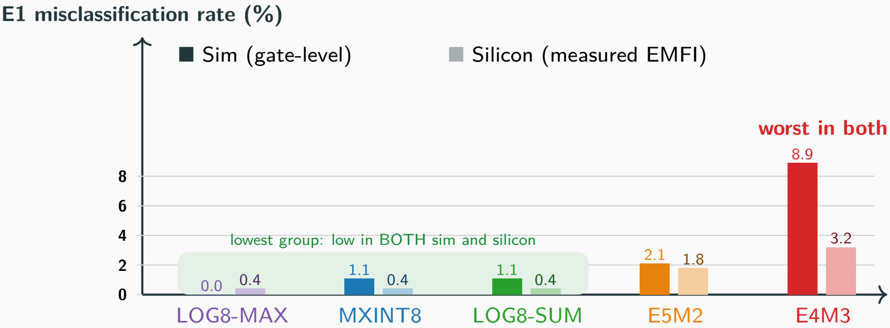
- Packaged **TSMC 180 nm** chip (RV32IC, **1 KB DFF SRAM**).
- EMFI coil, ± 20 V, 4 to 8 ns pulses.

No single format wins everywhere



$\Rightarrow$ Resilience is **format** $\times$ **workload**, not the format alone.

Case study (E1): simulation predicts the silicon ranking



Predicted containment matches measurement within **2.3%** across all five formats.

Conclusion: format choice is a reliability parameter

1. **Arithmetic structure, not bit-width alone, is a reliability parameter** in MX accelerator design.
2. Three MX-specific mechanisms drive the 8$\times$ gap:
   - ❶ SE amplification
   - ❷ routing inversion
   - ❸ self-healing
3. **No single format dominates:** MXINT8 is resilient at full accuracy; LOG8-MAX is fastest and most resilient where its $\sim$50% accuracy cap on multi-token attention is acceptable; E4M3 is the most exposed.

$\boxed{\textbf{Pick the MX format for resilience too, not accuracy and energy alone.}}$

Read our paper

Paper: `dl.acm.org/doi/10.1145/3787109.3815291`

- Detailed analysis of faults and vulnerabilities
- Additional experimental results
- Related work discussion
- Artifact: `github.com/Secure-Embedded-Systems/capri1_mx_faultanalysis`

Thank you! Questions?

dl.acm.org/doi/10.1145/3787109.3815291

github.com/Secure-Embedded-Systems/capri1_mx_faultanalysis

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This work vs. prior fault studies

Prior work	Formats	Shared exp?	Platform	Faults
Rakin '19, Yao '20, Hong '19	FP32	✗	Sim / DRAM	Bit-flip / RH
Breier '18, '26	INT8 / 4 std	✗	FPGA / SRAM	Laser, EMFI
MXDOTP '25, VMXDOTP '26	MX hardware	✓	RISC-V ext	*not studied*
Cuyckens '25, LNS-Madam '22	MX / log / CIM	✗	Sim	ECC only
This work	**5 MX 8-bit**	✓	**ASIC + GLS**	**Exhaustive 1-bit**

Prior MX work builds hardware but never injects faults; prior fault work never targets MX.

We are the first to do both.